SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

Code of Conduct:

I S.JAGADESH (*Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

CONCEPT MAPPING

1. A partially completed concept map is given below:

Application → Presentation → ______ → Transport → Network → ______ → Physical.

Analyze the missing layers and explain how their functions are essential for reliable communication.

2. Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.

3.Given the concept map:

DataLink→MACAddress→LocalDelivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.

4. A concept map shows:

ErrorControl

→DataLinkLayer→ErrorDetection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

5. Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.

6. A network issue is represented in the concept map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Concept Identification	20%	Accurately identifies all relevant OSI layers, concepts, and relationships	Identifies most concepts with minor omissions	Identifies limited concepts; some inaccuracies	Unable to identify key concepts correctly
Concept Mapping Structure	25%	Constructs a clear, logical, and well organized concept map with proper hierarchy and connections	Concept map is mostly clear with minor structural issues	Concept map lacks clarity, organization, or proper flow	No clear structure or incorrect mapping

Linking & Relationships	20%	Clearly establishes correct relationships between layers, functions, and processes	Relationships are mostly correct with minor errors	Limited or partially correct relationships	Incorrect or missing relationships
Application & Analysis	20%	Effectively applies concepts to explain processes (e.g., encapsulation, error control) with strong reasoning	The application is correct but lacks depth in analysis	Basic application with weak explanation	Unable to apply concepts correctly
Explanation & Clarity	15%	Provides a clear, concise, and well structured explanation supporting the concept map	Explanation is understandable with minor gaps	Explanation is unclear or incomplete	No clear or meaningful explanation

1. **Missing OSI Layers**

Concept Map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Answer:

- **Session Layer:** Establishes, manages, and terminates communication sessions.
- **Data Link Layer:** Provides framing, error detection, and reliable node-to-node communication.

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2. Data Units and OSI Layers

Concept Map:

- **Transport Layer → Segment**
- **Network Layer → Packet**
- **Data Link Layer → Frame**
- **Physical Layer → Bits**

Analysis:

Data changes from **Segment → Packet → Frame → Bits** as it moves down the OSI layers, enabling successful transmission over the network.

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3. MAC Address and IP Address

Concept Map:

- **Data Link → MAC Address → Local Delivery**
- **Network → IP Address → Global Delivery**

Answer:

- **MAC Address** delivers data within the local network.
- **IP Address** routes data between different networks.
- Together, they ensure complete end-to-end communication.

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4. Error Control

Concept Map:

- **Data Link Layer → Error Detection**
- **Transport Layer → Error Correction**

Analysis:

The Data Link Layer detects transmission errors, while the Transport Layer retransmits lost or corrupted data, ensuring reliable communication.

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5. Web Browsing Using OSI Layers

Concept Map:

Application → Presentation → Session → Transport → Network → Data Link → Physical

Analysis:

Each layer performs a specific task. If one layer fails (e.g., Network Layer), web pages cannot be delivered, causing communication failure.

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6. Network Troubleshooting

Concept Map:

Physical ✓ → Data Link ✓ → Network ✕ → Transport ✕ → Application ✕

Analysis:

The problem is at the **Network Layer**. Since routing fails, the Transport and Application layers cannot communicate. The concept map helps identify the faulty layer quickly.